INTERNAL ANALYSIS REPORT

Telecom Customer Churn Analysis

Understanding Why Customers Leave — and What We Can Do About It

This report presents a full analysis of customer churn across a telecom dataset of 7,043 customers. It covers the goals of the study, a chart-by-chart walkthrough of what the data revealed, a summary of key findings, and a set of actionable recommendations for the business team.

7,043	26.54%	Month-to-Month	Electronic-Check
Total Customers Analysis	Overall Churn Rate	Highest Risk Contract Type	Highest Risk Payment Method

Prepared by: Mrinmoy Talukdar

| Dataset: 7,043 Customers | April 2025

01 | What Are We Trying to Do?

Project Objective

Customer churn is one of the most expensive problems a telecom company faces. Every time a customer decides to leave, we don't just lose their monthly revenue — we lose everything we invested in acquiring them in the first place. And in a market where competition is fierce, getting new customers to replace the ones we lose is getting harder and more costly every year.

So the question we set out to answer is simple but important: **Who is leaving, and why?** If we can figure out the patterns behind churn, we can take action before it happens — not after.

Specific Goals

- Measure the actual scale of churn — how many customers are we losing, and what percentage does that represent?
- Identify which customer groups are churning at higher rates — seniors, short-tenure users, specific contract types.
- Understand which services and payment methods are linked to higher churn.
- Translate data findings into clear, actionable recommendations for the business team.

This is not just a numbers exercise. The goal is to give the business team a clear picture of where customers are slipping away — and hand them concrete levers to pull.

The Dataset

We worked with a dataset of 7,043 telecom customers. Each record captures demographic info, the services they subscribed to, their contract type, payment method, monthly charges, and — most importantly — whether they churned (left the service) or stayed. After cleaning the data (fixing a few labelling errors and converting blank charge fields to zero for new customers), we had a clean, complete dataset with no missing values and no duplicate entries.

Column	What It Tells Us
gender, Senior Citizen	Demographic profile of the customer
Partner, Dependents	Whether they live with a partner or have dependents
tenure	How many months they've been a customer
PhoneService, InternetService, …	Which services they subscribed to
Contract	Month-to-month, one year, or two year
PaymentMethod	How they pay their bills
MonthlyCharges, TotalCharges	Revenue metrics
Churn (target)	Yes = left, No = stayed

02 | What the Data Showed Us — Chart by Chart

Below, we walk through each chart from our analysis. Every graph sits on its own page with a plain-language explanation of what it shows and why it matters. No jargon, no fluff.

Chart 1 — Churn Count: How Many Customers Actually Left?

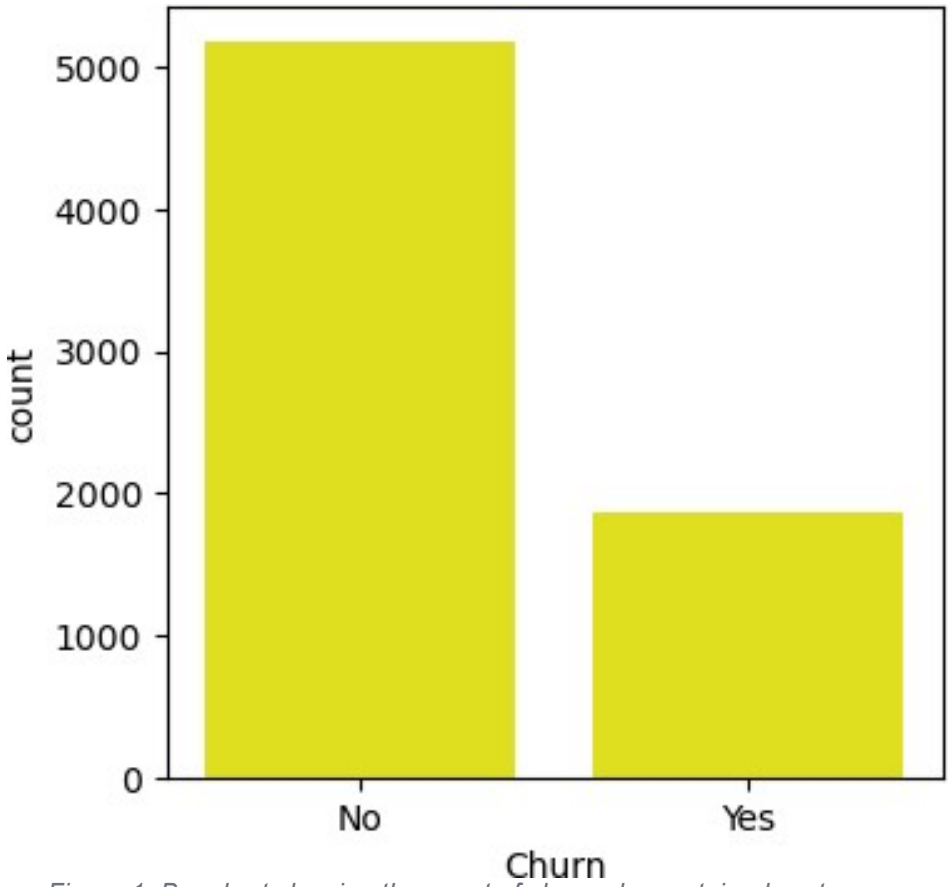

Figure 1: Bar chart showing the count of churned vs. retained customers

Before we could dig into the 'why', we needed to understand the 'how many'. This first chart gives us the raw count — how many customers stayed versus how many left during the period we analysed.

Looking at the chart, there's a clear imbalance between the two bars. The 'No' bar (customers who stayed) is noticeably taller than the 'Yes' bar (customers who left). That's somewhat expected — most businesses retain the majority of their customers at any given time. But the gap isn't as wide as you'd hope to see.

Key Takeaways

- The volume of churned customers is large enough to be a serious business problem — not a rounding error.
- This count forms the baseline: every chart that follows helps explain what's driving these numbers.

Chart 2 — Exact Churn Numbers with Labels

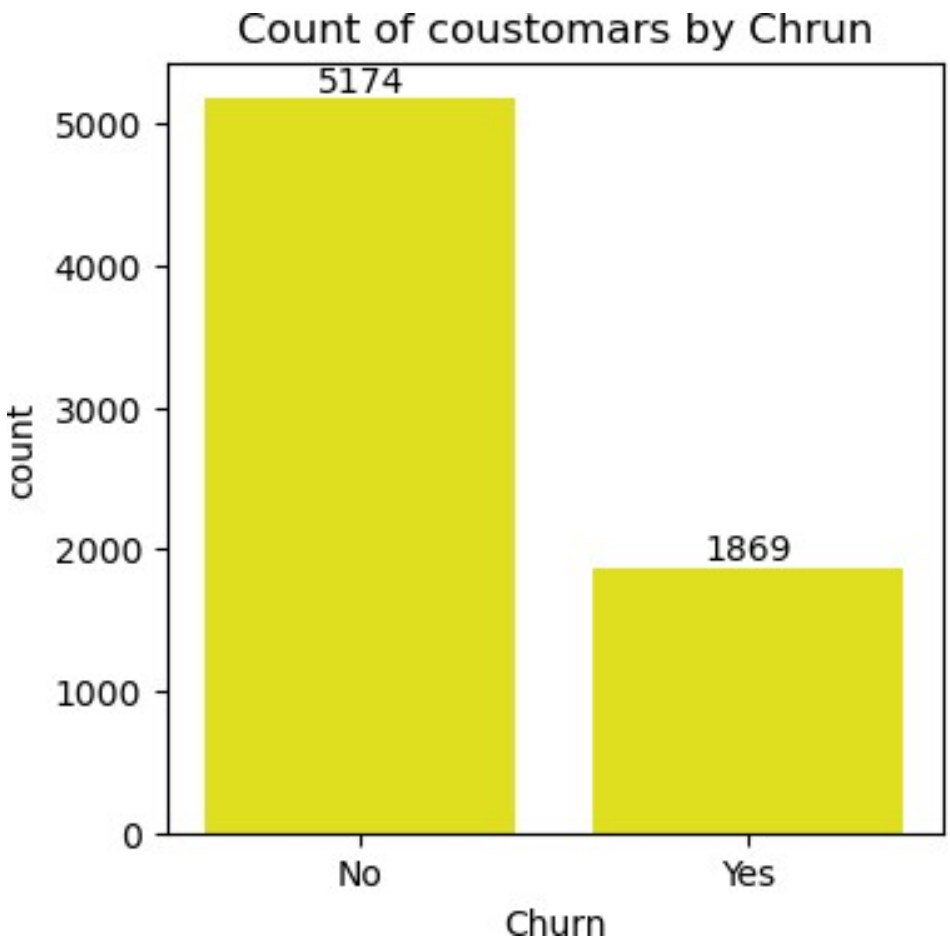

Figure 2: Count of customers by churn status with exact values on each bar

This is the same chart as before, but now with the exact numbers sitting on top of each bar. Numbers on their own can be a bit abstract, so seeing them written right there on the chart makes the comparison much easier to process at a glance.

The labelled version shows us precisely how many customers fall into each group. When you're presenting findings to stakeholders, having these exact figures visible removes any ambiguity about what we're working with.

Key Takeaways

- Exact figures matter when communicating to business teams — approximations lead to underestimation of the problem.
- These numbers set the stage for the percentage-based analysis in the next chart.

Chart 3 — Churn Rate: 26.54% of Customers Have Left

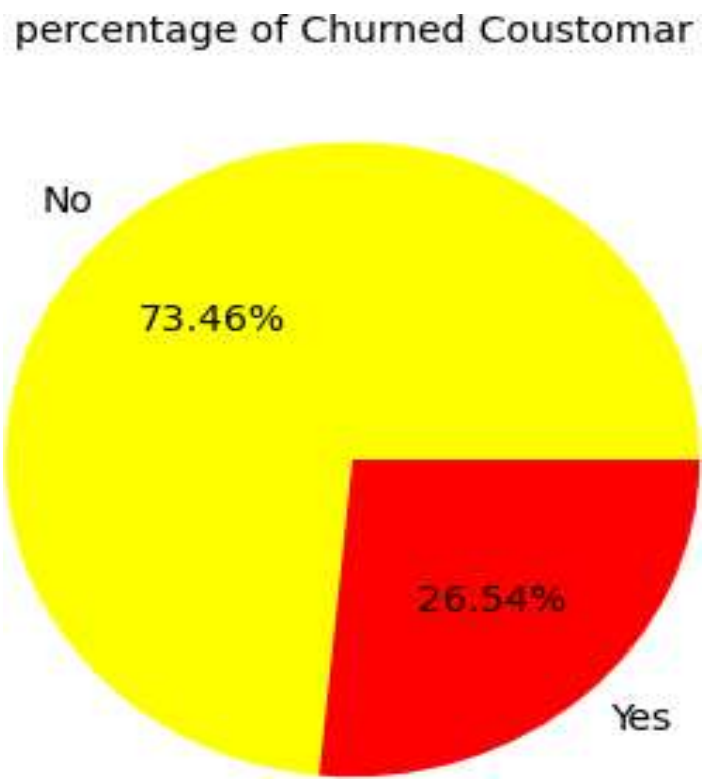

Figure 3: Pie chart showing the percentage split between churned and retained customers

Percentages put things in perspective in a way that raw counts sometimes don't. This pie chart tells us that **26.54% of our customer base has churned** — that means roughly 1 in every 4 customers is walking away.

To put that in context: if you imagine a room of 100 customers, 27 of them have already decided to leave. That's not a small leak — that's a significant drain. Industry benchmarks for telecom churn typically sit around 15–25%, so our figure is at the higher end of what's considered normal, which makes it even more important to understand and address.

The red slice representing churned customers is visually striking against the yellow. It's big enough that no one looking at this chart can argue this isn't a priority issue.

Key Takeaways

- A 26.54% churn rate places us at the upper range of typical telecom churn — urgent but fixable.
- Nearly 1 in 4 customers is leaving, which translates to a major revenue recovery opportunity.

Chart 4 — Does Gender Affect Churn?

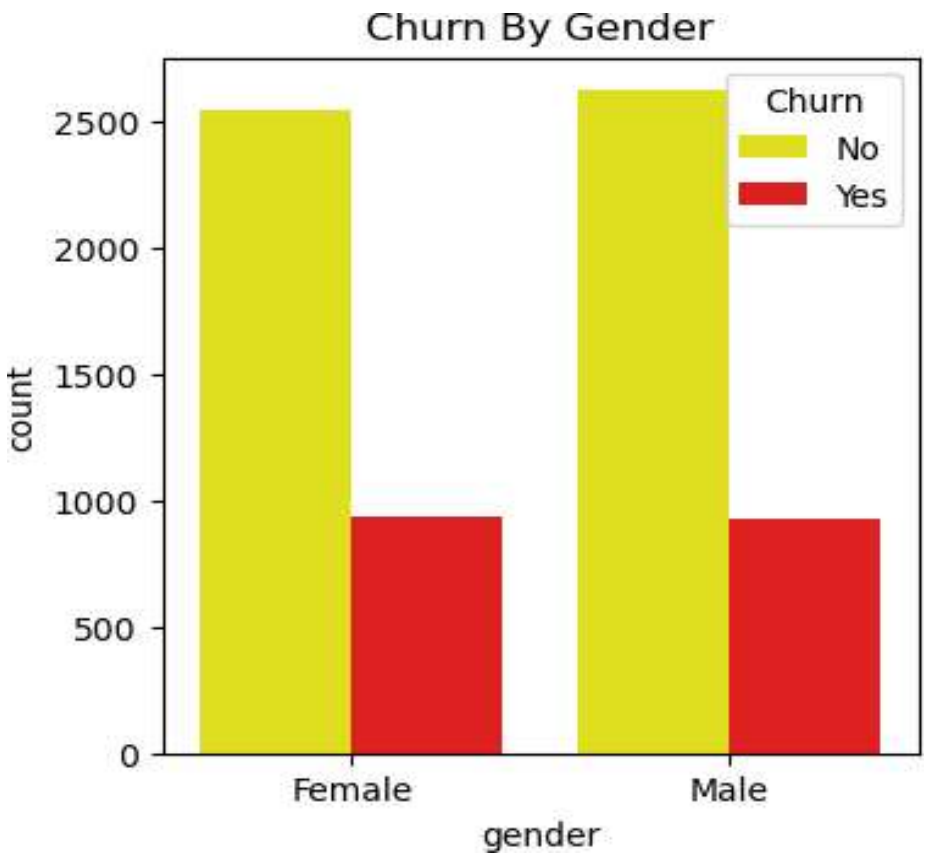

Figure 4: Churn comparison across male and female customers

One of the first things a business team asks when they see churn data is: 'Is this affecting one group more than another?' So we looked at gender — are male customers more likely to leave than female customers, or vice versa?

The chart shows something quite clear: the churn pattern across genders is almost identical. Both male and female customers are churning at roughly the same rate. The bars are nearly the same height on both sides, and the proportion of red (churned) to yellow (retained) is consistent.

This tells us that gender is **not a meaningful driver of churn** in this dataset. It's a useful finding because it lets us rule out one variable and focus our attention elsewhere.

Key Takeaways

- Gender is not a churn risk factor — targeting by gender would be a wasted effort.
- Marketing and retention strategies should be built around other, more predictive variables.

Chart 5 — Senior Citizens in Our Customer Base

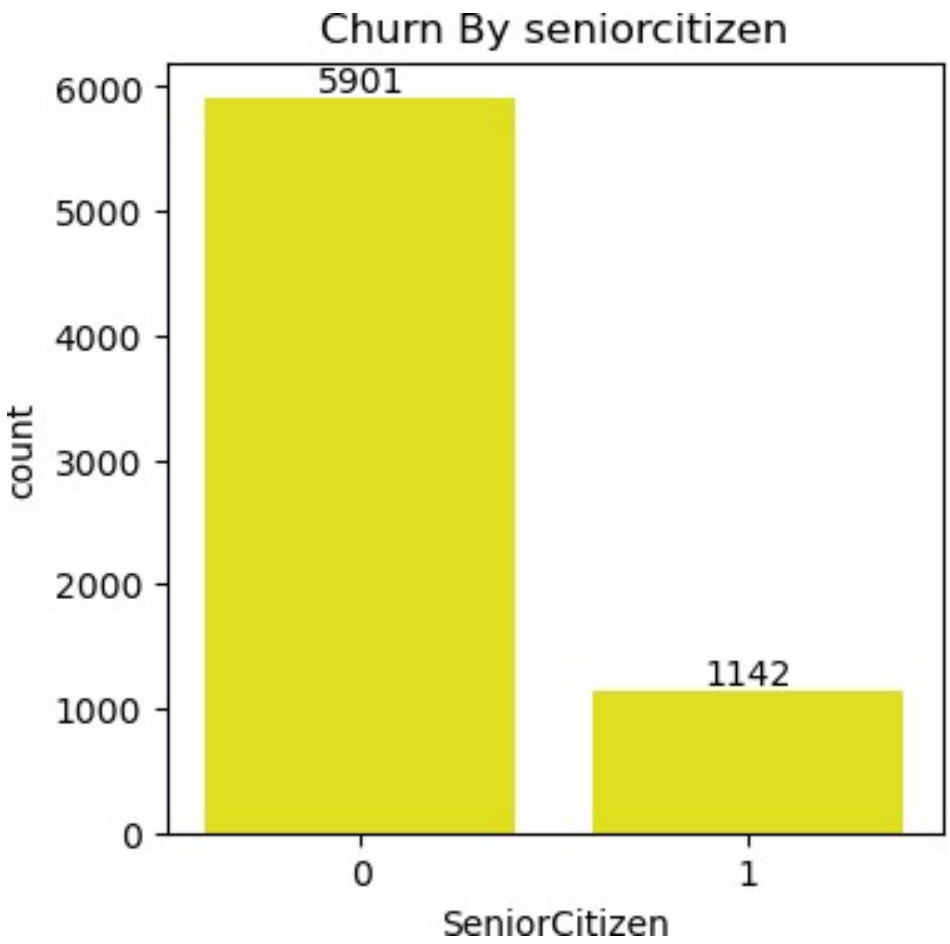

Figure 5: Count of senior vs. non-senior customers

Before we could understand whether senior citizens churn more, we first needed to know how many of our customers actually fall into that category. This chart does exactly that — it shows the distribution of senior versus non-senior customers in our base.

The chart reveals a significant skew: the vast majority of our customers are **not** senior citizens. Senior customers make up a much smaller portion of the overall base. This matters because even if their churn rate is higher, the absolute number of seniors churning will be smaller than the churn coming from the non-senior majority.

But don't let the smaller count fool you — the churn rate within that group, as the next chart will show, is disproportionately high.

Key Takeaways

- Senior citizens are a minority segment, but one that deserves dedicated attention.
- Smaller group size + higher churn rate = a concentrated problem worth solving specifically.

Chart 6 — Senior Citizens Churn at a Much Higher Rate

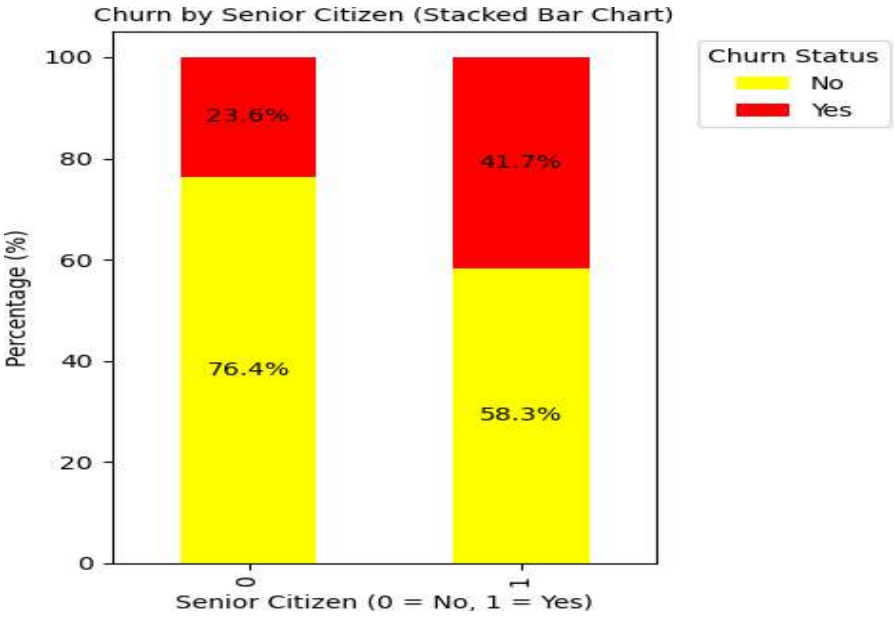

Figure 6: Stacked bar chart showing churn percentage by senior citizen status

This is where things get really interesting. The stacked bar chart breaks down the churn rate *within* each group — seniors versus non-seniors — as a percentage. This removes the distortion caused by the different group sizes and lets us compare apples to apples.

The chart makes the disparity unmistakable. A **significantly higher percentage of senior citizens are churning** compared to the non-senior group. While the non-senior bar shows a relatively small red section (churned), the senior bar has a much larger red portion.

What might be driving this? Senior customers may have different service needs, may find the technology harder to navigate, may be more sensitive to price increases, or may simply not feel supported enough. Whatever the root cause, this group is telling us — through their departure — that something isn't working for them.

Key Takeaways

- Senior citizens churn at a disproportionately high rate — they need targeted support and engagement.
- Consider dedicated account management, simplified plans, or senior-friendly onboarding programs.
- Proactive outreach to senior customers could significantly reduce churn in this segment.

Chart 7 — New Customers Are Leaving the Fastest

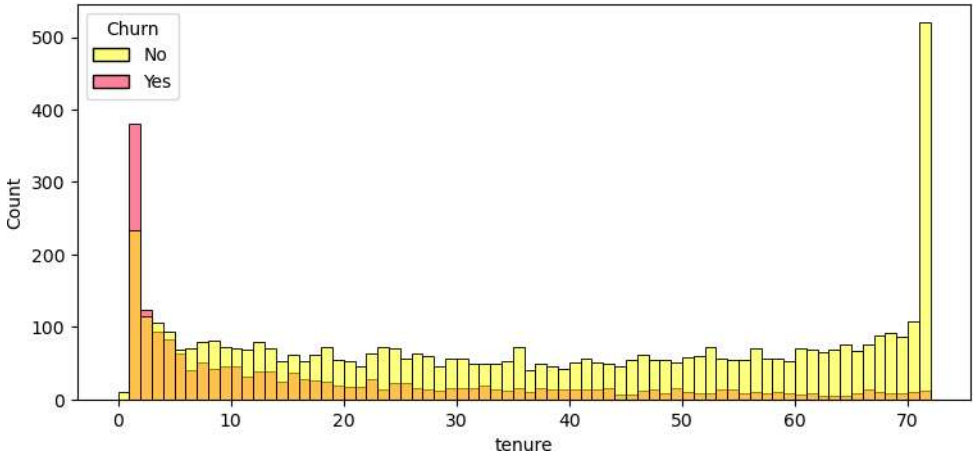

Figure 7: Histogram of customer tenure split by churn status

How long has a customer been with us? This histogram plots tenure — the number of months a customer has been using our service — and colours each bar by whether that group churned or stayed.

The pattern that jumps out immediately is at the left end of the chart — the very first few months. The red bars (churned customers) are tallest right at the beginning of the customer journey. In other words, **customers who leave tend to leave early**. They sign up, and within the first month or two, they decide it's not for them.

Contrast this with the right side of the chart. Customers who have been with us for 50, 60, or 70+ months are almost entirely represented by yellow bars — they stay. Long-term customers are loyal customers. The longer someone stays, the less likely they are to leave.

This is a classic 'early churn' problem, and it has a clear implication: **the onboarding experience is where we're losing people**. If we can get customers through the first 3–6 months with a positive experience, the data suggests they're far more likely to stick around.

Key Takeaways

- The highest churn risk is in the first 1–3 months of a customer's journey.
- Long-tenure customers are significantly more loyal — retention compounds over time.
- Investing in onboarding quality, early check-ins, and first-month experience will have the biggest impact on churn reduction.

Chart 8 — Month-to-Month Contracts Are a Churn Hotspot

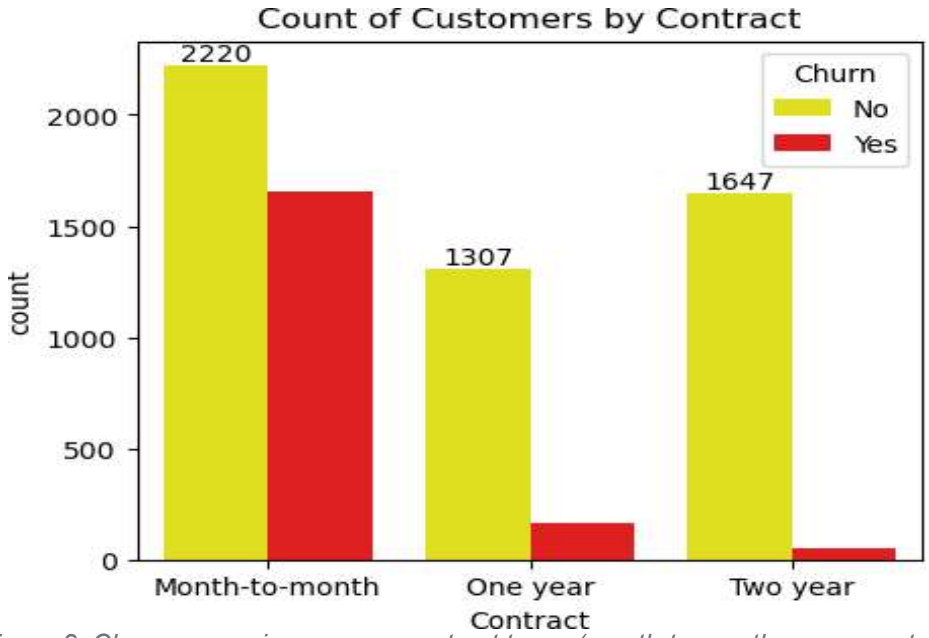

Figure 8: Churn comparison across contract types (month-to-month, one year, two year)

Contract type turns out to be one of the most powerful predictors of churn in this entire dataset. This chart compares three groups: customers on month-to-month contracts, one-year contracts, and two-year contracts.

The difference is stark. Customers on **month-to-month contracts have an overwhelmingly higher churn rate** than those on longer-term agreements. The red bar for month-to-month is dramatically taller than the red bars for one-year or two-year contracts.

This makes intuitive sense. When a customer is on a month-to-month plan, there's no contractual friction holding them in place. If they're unhappy for any reason — price, service quality, a competitor's offer — they can leave without penalty. Annual and two-year customers, on the other hand, have made a longer commitment and are less likely to disrupt it.

The business implication here is direct: **we need to actively encourage customers to move from month-to-month to longer contracts**. Incentives like discounts, free months, or added benefits for signing a one-year or two-year deal could have a meaningful impact on churn numbers.

Key Takeaways

- Month-to-month customers are far more likely to churn than those on annual or two-year plans.
- Offering incentives to upgrade contract length (discounts, loyalty perks) is a high-ROI retention strategy.
- Consider making longer contracts more attractive upfront rather than waiting for customers to consider leaving.

Chart 9 — Add-On Services Reduce Churn Significantly

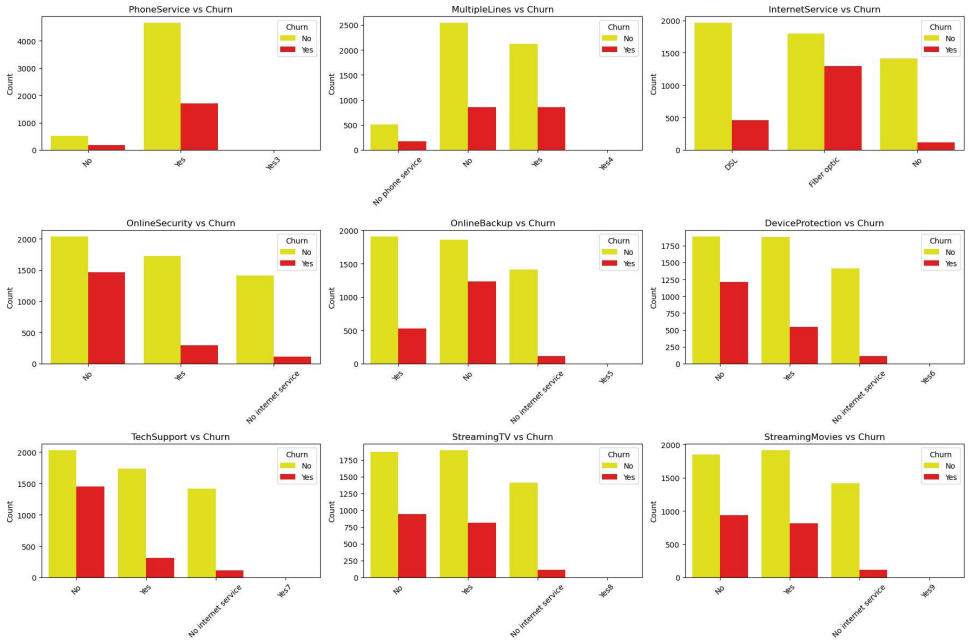

Figure 9: Churn breakdown across 9 additional services (online security, backup, tech support, streaming, etc.)

This chart is a grid showing churn rates across nine different service add-ons: phone service, multiple lines, internet service type, online security, online backup, device protection, tech support, streaming TV, and streaming movies.

A consistent pattern emerges across almost every service: **customers who do not subscribe to a service churn at a higher rate than those who do**. This is particularly visible for services like online security, online backup, tech support, and device protection — where the red bars for 'No' (not subscribed) are noticeably taller.

There's also something interesting about internet service type. Fiber optic customers — who typically pay more — are churning at a higher rate than DSL customers. This may suggest that fiber optic customers have higher expectations, or that the price premium isn't feeling worth it to them.

The broader takeaway here is that **service engagement is a retention signal**. Customers who use more services are more embedded in the ecosystem and less likely to leave. Getting customers set up with additional services early on isn't just a revenue play — it's a retention strategy.

Key Takeaways

- Customers without add-on services like security, backup, and tech support churn at a higher rate.
- Service engagement = customer stickiness. The more services they use, the less likely they are to leave.
- Fiber optic customers need particular attention — their churn rate is higher, possibly due to unmet expectations.
- Bundling or offering free trials of protective services (security, backup, tech support) could boost retention.

Chart 10 — Electronic Check Users Are Churning at Worrying Rates

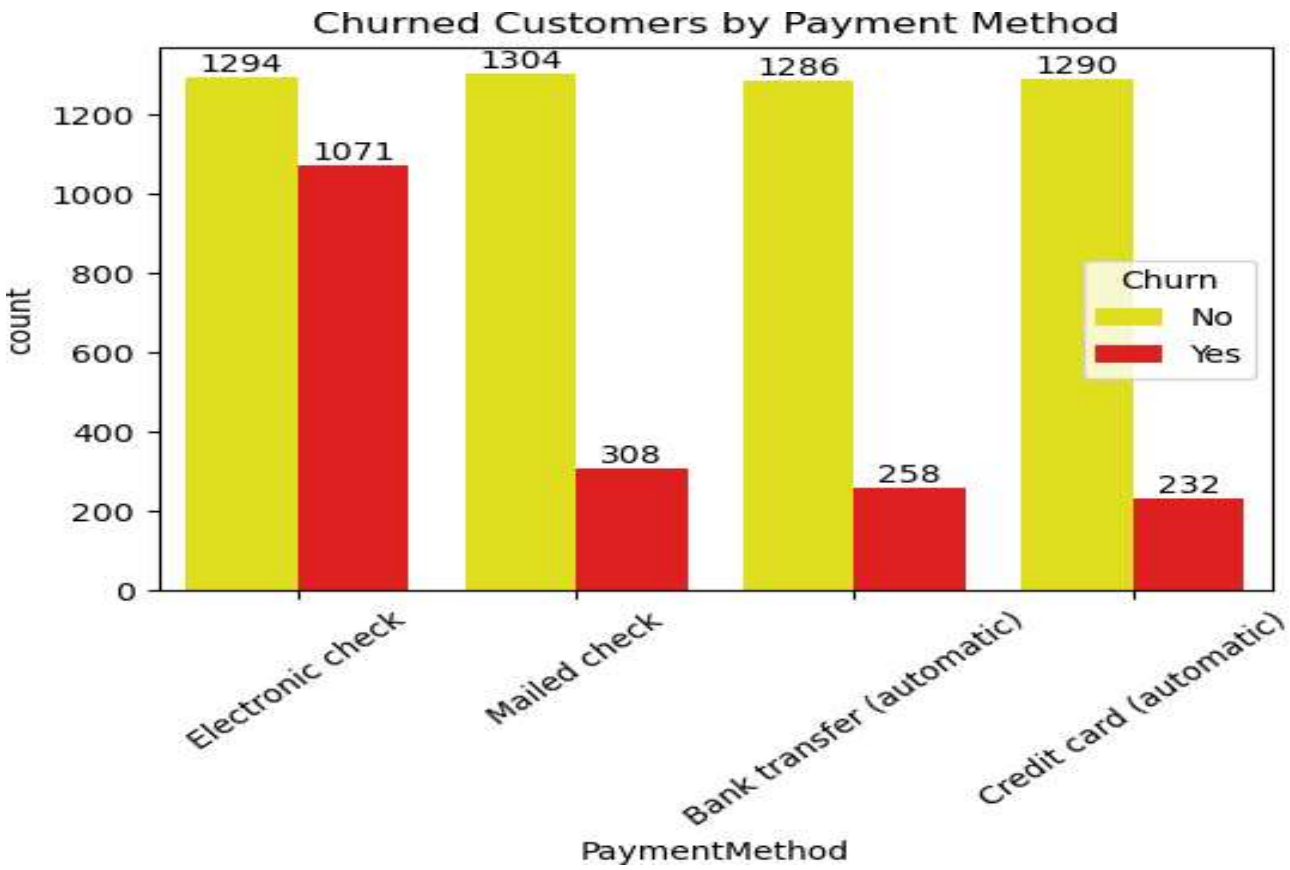

Figure 10: Churn breakdown by payment method (electronic check, mailed check, bank transfer, credit card)

Our final chart looks at payment method — something that might not seem immediately related to churn, but the data tells a different story.
There are four payment methods in the dataset: electronic check, mailed check, bank transfer (automatic), and credit card (automatic). The chart shows churn rates for each, and the differences are significant.

Electronic check users have by far the highest churn rate. Mailed check users also churn at a relatively higher rate. In contrast, customers who pay via bank transfer or credit card (both automatic methods) are much more likely to stay.

Why might this be? Automatic payment methods create a passive commitment — the bill is paid without the customer having to actively do anything, which reduces the monthly 'friction moment' where they might reconsider their subscription. Manual payment methods (like writing a cheque or initiating an electronic payment) force the customer to consciously engage with the bill every month, which is a more active decision point where they might choose to stop.

There's also a demographic angle worth exploring: electronic check users may be a segment that's less tech-engaged or more price-sensitive — factors that could compound the churn risk.

Key Takeaways

- Electronic check users represent the highest churn risk by payment method — by a clear margin.
- Automatic payment methods (bank transfer, credit card) are associated with significantly lower churn.
- Incentivizing customers to switch to automatic payments could reduce churn passively.
- Consider targeted outreach to electronic check and mailed check users offering payment method switch incentives.

Summary of Key Findings

After going through all ten charts and the patterns they revealed, here's the honest summary of what the data is telling us:

26.54% churn rate
More than 1 in 4 customers is leaving. This is at the upper end of typical telecom churn and represents a material revenue risk.

Gender doesn't matter here
Male and female customers churn at almost identical rates. Gender-based targeting won't help.

Senior citizens are at much higher risk
Despite being a smaller segment, senior customers are churning disproportionately. They likely need more support and tailored engagement.

New customers are the most vulnerable
The bulk of churn happens in the first few months. If a customer survives past month 6, they're far more likely to stay long-term.

Month-to-month contracts are the churn engine
Customers with no long-term commitment churn far more than those on annual or two-year plans. This is the most controllable variable we identified.

Add-on services act as a retention anchor
Customers who use online security, backup, tech support, and other services are more loyal. More services = more stickiness.

Fiber optic users need attention
Higher-paying fiber customers are churning more than DSL users — possibly a value perception problem.

Electronic check users are a high-risk group
Manual payment users, especially electronic check payers, churn significantly more than automatic payment users.

Actionable Recommendations

These aren't generic suggestions pulled from a textbook. Each recommendation below comes directly from a specific finding in the data. The more of these we act on, the more we expect churn to come down.

1. Fix the first 90 days	Since most churn happens right at the start, we need to rethink what the first 1–3 months look like for a new customer. This means a better onboarding process, proactive check-in calls or messages after the first 30 days, and making sure customers understand and use the services they've signed up for.
2. Push longer contracts with real incentives	Month-to-month plans are convenient for customers but dangerous for us. We should offer meaningful discounts or added benefits to customers who switch to annual or two-year plans. Even moving 10–15% of month-to-month customers to an annual plan could have a significant impact on churn.
3. Build a senior citizen support programme	Senior customers are leaving at higher rates. A dedicated support tier — simpler plans, more patient customer service, proactive check-ins, clear billing — could make a real difference. This segment may also respond well to loyalty rewards for long tenure.
4. Bundle and promote protective services	Online security, backup, and tech support aren't just add-ons — they're retention tools. We should make it easier and cheaper for customers to access these services, possibly offering free trials in the first month or bundling them into entry-level plans. Engaged customers stay.
5. Nudge electronic check users toward automatic payments	A targeted campaign to move electronic check and mailed check users to automatic payment methods could reduce churn passively. Offer a small monthly discount or a bill credit for switching — it costs less than the revenue lost to churn.

6. Investigate the fiber optic problem	Higher-paying fiber customers churning more than DSL users is a red flag. We need to find out whether this is a price perception issue, a service quality issue, or an expectation mismatch — and address it specifically. A brief survey to churned fiber users would give us direct answers.

Churn isn't inevitable. The data shows us exactly where the problem is concentrated. If we act on even two or three of these recommendations, we expect a measurable reduction in churn — and a meaningful recovery in annual revenue.

This report was prepared by the Data Analytics team using Python (pandas, seaborn, matplotlib). The dataset covers 7,043 customers with 21 variables. All cleaning steps (label correction, type conversion, null checks, duplicate verification) were completed before analysis. Charts reproduced directly from the analysis notebook.